

Quiz 5

Problem Solving, Math 1000 – Summer 2016

Wednesday, June 14

The following quiz is out of 60 points, and will count towards your quiz grade. Answer the following problems (you may use your notes).

Problem 1 Find the APY for an APR of 7.7% compounded:

(a) *annually*

7.7%

(b) *semi-annually (twice a year)*

7.85%

(c) *monthly*

7.98%

(d) *continuously*

8%

Problem 2 Consider a CD paying a 8% APR compounded continuously. Find the future value of the CD if you invest \$4,000 for a term of 4 years.

The future value of the CD if you invest \$4000 for a term of 4 years is \$5508.51.

Problem 3 Between the years 1994 and 2004, the average annual inflation rate was 3.7%. Find the salary in the year 2004 that would be equivalent to a \$50,000 salary in 1994.

The salary in 2004 dollars that would be equivalent to a \$50,000 salary in 1994 is \$71,904.75.

Problem 4 Consider a CD paying a 4.9% APR compounded daily.

(a) Find the periodic interest rate.

The periodic interest rate is 0.013%.

(b) Find the future value of the CD if you invest \$3,000 for a term of five years.

The future value of the CD is \$3,803.22.

Problem 5 Let's say on January 01, 2016 you are buying a house for \$200,000 (after down payment, realtor fee, title fee, points fee, etc.). You get a loan for 30 years at 3% APR compounded monthly, and \$843.21 is your monthly payment.

(a) How much interest in total will you end up paying on this loan?

$$T = 30 \times 12 = 360 \text{ months}$$

$$I = M \times T - P$$

$$\$103,555.60$$

(b) Suppose you get paid every two weeks, and decide to pay \$421.61 each payday instead (and have the compounding period adjusted accordingly). When is the loan paid off?

$$M = P \frac{p(1+p)^T}{(1+p)^T - 1}$$

$$M = \$421.61$$

$$P = \$200,000$$

$$p = \frac{0.03}{26}$$

$$T = ?$$

$$26 \text{ years } 24 \text{ weeks later } (T = 688 \text{ biweekly periods})$$

(c) How much interest in total will you end up paying with the biweekly format?

$$I = M \times T - P$$

$$\$90,067.68$$